

EDAA75 Exam

5 June 2025

Instructions

Things you CAN use during the exam.

Any written or printed material is fine. Textbook, other books, the printed slides, handwritten notes, whatever you like.

In any case, it would be good to have a source for the relevant definitions, and also for notation, just in case you don't remember the precise definition of everything we discussed in the course.

Things you CANNOT use during the exam.

Anything electrical or electronic, any communication device: computers, calculators, mobile phones, toasters, ...

A sheet with common symbols and notations is attached at the end.

Good luck!

1	2	3	4	5	total
22	16	16	26	20	100

Total points: 100
points required for 3: 50
points required for 4: 67
points required for 5: 85

In questions **1** and **2** below (pages 2 and 3), you don't need to justify your answers unless explicitly asked to.

1**[22 pt]**

1. [10pt] Classify each of the following binary relations as

- **E**: An equivalence relation.
- **T**: A total order.
- **P**: A partial order that is not total.
- **S**: A symmetric relation that is not transitive.
- **N**: None of the above.

(a) The relation xRy on $\mathbb{Z}$ such that $x - y$ is divisible by 5. _____

(b) The relation xRy on $\mathbb{Z}$ such that $x < y$. _____

(c) The relation xRy on the set $\{1, 2, 3, 4, 5\}$ such that $|x - y| \leq 1$.¹ _____

(d) The relation xRy on $\mathcal{P}(\{1, 2, 3\})$ such that $x \subseteq y$.² _____

(e) The relation xRy on $\mathbb{Z}$ such that $|x - y| = 2$.¹ _____

2. [12pt] Formulas defining functions from integers to integers are listed below. For each function, indicate whether it is:

- **B**, a bijection
- **S**, a surjection, but not a bijection,
- **I**, an injection, but not a bijection,
- **N**, neither an injection nor a surjection.

(a) $a(x) := x^2$ _____

(b) $b(x) := x + 2$ _____

(c) $c(x) := 2x$ _____

(d) $d(x) := -x$ _____

(e) $e(x) := \lfloor x/2 \rfloor$,
that is, the quotient of x divided by 2 rounded down to the next integer. _____

(f) $f(x) := (c \circ e)(x)$, that is, the composition of functions c and e defined above. _____

¹Here $|x - y|$ denotes the absolute value of $x - y$.

²For a set S , the set $\mathcal{P}(S)$ denotes the power set of S .

2**[16 pt]**

Given sets $A, B \subseteq \mathbb{Z}$, let the set $P_{A,B} \subseteq \mathbb{Z}$ be the set that contains all products of an element in A with an element in B . For example, if $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, then $P_{A,B} = \{4, 5, 6, 8, 10, 12, 12, 15, 18\}$.

1. [2pt] Given sets $A, B \subseteq \mathbb{Z}$, define the set $P_{A,B}$ using set builder notation.

$$P_{A,B} :=$$

2. [2pt] True or false: $\forall A, B \subseteq \mathbb{Z}, \forall x \in P_{A,B}, \forall y \in B (\neg(\forall z \in A (z \cdot y \neq x)))$ _____

3. [2pt] Suppose $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$. What is the size of $P_{A,B}$? _____

4. [10pt] Suppose $A = \{2, 3\}$ and $B = \mathbb{N}$.

(a) Is there a surjective function from $P_{A,B}$ to $\mathbb{Z}$? If so, define one; if not, explain why not.

(b) Is there a bijection between $P_{A,B}$ and $\mathbb{Z}$? Justify your answer.

3**[16 pt]**

Solve either **OPTION A** or **OPTION B** below (not both).

OPTION A: Number theory

1. [6pt] Prove that an integer is divisible by 4 if and only if the number formed by its two rightmost digits (in base 10) is divisible by 4.

3. [10pt] In class we proved the following theorem.

Theorem 1. *The greatest common divisor of a and b is a linear combination of a and b . That is,*

$$\gcd(a, b) = sa + tb$$

for some integers s and t .

For positive integers a, b, c , state whether the following statements are TRUE or FALSE and then prove or give a counter-example.

You may not appeal to uniqueness of prime factorization (because some of these properties are needed to prove unique factorization) but you can use the theorem above.

- (a) $c \mid a$ and $a \mid b$ if and only if $c \mid \gcd(a, b)$
- (b) If $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$, then $\gcd(a, bc) = 1$

OPTION B

1. [8pt] Rewrite the following logical formula in prenex form. Pay attention to variable scope.

$$\neg \forall x. (\exists y. (P(x, y) \rightarrow \forall z. Q(y, z)) \vee (\exists y. R(y, z)))$$

2. [8pt] For each definition of predicates p and q in the items below, answer whether the statement is:

- **E**: p is equivalent to q
- **I**: p implies q but q does not imply p
- **R**: q implies p but p does not imply q
- **N**: none of the above

Either explain your reasoning step by step or draw a truth table (on another sheet) to illustrate your reasoning. You may use as many columns as you need.

(a) p is $(r \vee s) \rightarrow t$ and q is $(\neg t \rightarrow \neg r) \vee (\neg r \rightarrow \neg s)$

(b) p is $(r \rightarrow s) \rightarrow t$ and q is $\neg r \vee \neg s \vee t$

4**[26 p]**

Let $G = (V, E)$ be a directed graph with no self-loops. We say a vertex $y \in V$ is reachable from a vertex $x \in V$ if there exists a directed path from x to y .³ We say vertices x and y are *strongly connected* if x is reachable from y and y is reachable from x . We say G is strongly connected if every vertex is reachable from every other vertex. Let $P(x, y)$ be the predicate “there exists a directed path from x to y ”.

This task consists of two parts. For the first part, your task is to express other predicates about the graph G in terms of $P(x, y)$. In the second part, you are asked to (recursively) define $P(x, y)$.

PART I [12pt]

Express each of the following sentences in terms of P , quantifiers, logical connectives, and equality, using variables that range over the vertices of G .

1. Vertices x and y are strongly connected.

2. G is not strongly connected.

3. G has a vertex of outdegree zero. (Reminder: we are considering graphs that only have edges between distinct vertices, that is, no self-loops.)

4. There is a positive-length walk⁴ from x to y .

³Recall that a path cannot repeat vertices, but for every vertex x there is a length-0 path from x to itself.

⁴A walk is a path that can go through the same vertex more than once.

PART II [14pt]

Your goal is now to (recursively) define a function $P : V \times V \longrightarrow \{\text{TRUE}, \text{FALSE}\}$ that given two vertices x and y , outputs TRUE if there exists a directed path from x to y and FALSE otherwise.

In order to define $P(x, y)$, we first define a function $f : V \longrightarrow \mathcal{P}(V)$ that, given v as input, outputs all vertices that are reachable from v . We then define

$$P(x, y) = \begin{cases} \text{TRUE} & \text{if } y \in f(x), \\ \text{FALSE} & \text{otherwise.} \end{cases}$$

Consider the following (wrong) definition:

$$f : v \mapsto \{v\} \cup \bigcup_{w:(v,w) \in E} f(w).$$

- (a) What is wrong with this definition?

There are many ways to fix this definition. One way is to define an auxiliary function $g : V \times \mathcal{P}(V) \longrightarrow \mathcal{P}(V)$ that given a vertex $v \in V$ and a set of vertices $S \subseteq V$ outputs the set of all vertices $U \subseteq S$ that are reachable from v .

- (b) Define the function g recursively below and briefly explain why your definition does not have the same problem as the one in the wrong definition.

$$g : v, S \mapsto \left\{ \begin{array}{l} \end{array} \right.$$

- (c) Now how do we define f using g ?

5**[20 p]**1. [12pt] **Coloring forests**

Background: In a large corporation, employees are grouped under various independent departments. Each department has a strictly hierarchical reporting structure: each employee reports to exactly one manager, except the department head who reports to no one. No employee has more than one manager, and no circular reporting is allowed.

For an upcoming annual review, you want to assign one of k different interviewers to each employee. To avoid bias, you must ensure that no employee and their direct manager are assigned the same interviewer.

Problem:

- (a) Model this scenario as a graph coloring problem. Explain why proper coloring applies here, and why the graph structure is a forest (a collection of trees).
- (b) Assuming there is at least one employee with a direct manager, what is the smallest k so that we are guaranteed there exists such an assignment of interviewers to employees?
- (c) Suppose the forest (company-wide structure) has n employees and c departments, all of them non-empty. Prove *by induction* on n that the number of valid interviewer assignments (i.e., proper colorings using k colors) is:

$$k^c(k-1)^{n-c}.$$

2. [8pt] **Scheduling tasks**

Background: You are designing a task scheduling system where n tasks must be assigned to machines, with various constraints and priorities.

Problem:

- (a) How many different ways are there to assign n distinct tasks to **two identical machines**, where each task is assigned to exactly one machine and the order of execution on each machine matters? (Assume machine identity does not matter, but the order of tasks on each machine does.)
- (b) Suppose now that each task has one of three priority levels: **high**, **medium**, or **low**. Each machine must process high-priority tasks first, followed by medium-priority, then low-priority, and within each level, the order matters.

For **a single machine**, how many valid schedules are there for 12 distinct tasks where 4 are high priority, 5 are medium priority, and 3 are low priority?

Some common symbols

- $\mathbb{N}$ the natural numbers, starting at 0
- $\mathbb{N}^+$ the natural numbers, starting at 1
- $\mathbb{Z}$ the integers
- $\mathbb{R}$ the real numbers
- $\mathbb{Q}$ the rational numbers
- $a \perp b$ a and b are coprime, i.e., they do not have a common divisor other than 1
- $a \mid b$ a divides b , i.e., $\exists k \in \mathbb{Z}$ such that $ka = b$
- $\mathcal{P}(A)$ power set of A
- $\overline{R}$ of a relation R : its complement
- R^{-1} of a relation R : its inverse
- $R \circ S, f \circ g$ of relations and functions: their composition or a function f
- $[a, b],]a, b[,]a, b], [a, b[$ closed, open, and half-open intervals from a to b
- $A \sim B$ two sets A and B are equinumerous
- A^* for a finite set A , the set of all finite sequences of elements of A , including the empty sequence, ε
- $\sum S$ sum of all elements of S
- $\prod S$ product of all elements of S
- $\bigcup S$ union of all elements of S
- $\bigcap S$ intersection of all elements of S
- $\bigcup_{a \in S} E(a), \bigcap_{a \in S} E(a)$ generalized union / intersection of the sets computed for every a in S